

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I S. Shamah (192513002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Shamah
Signature of the student:

17
20

B

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

INTELLIGENT AGENTS IN SMART HOME AUTOMATION:

An Intelligent agent is a system that observes its environment through sensors and take actions to achieve a desired goal.

1. Simple Reflex Agent:

A simple reflex agent acts only according to the current situation using condition action rules.

- * If motion is detected $\rightarrow$ turn on the light.
- * If temperature is high $\rightarrow$ turn on the AC.
- * If no motion is detected $\rightarrow$ turn off the light.

2. Model-Based Agent:

A model based agent maintains an internal model of the environment. It uses both current information and previous information to make decisions.

3. Goal-Based Agent:

A goal-based agent ~~selects~~ actions to achieve specific goals such as:

- * Maintaining comfortable temperature.
- * Saving energy
- * Improving home security.

4. Utility-Based Agent:

* A utility-based agent chooses the action that gives the best overall result.

* It can balance comfort, energy consumption, and security.

* For example, it can decide whether to turn on the AC by considering temperature, time, energy usage, and user preferences.

Conclusion:

* A hybrid intelligent agent is suitable for smart homes because it can respond to real-time conditions and make better decisions based on user preferences and environmental information.

* It can save more time in house hold works and can be used efficiently in many ways.

* It can save much time and can be used in smart-homes to increase reliability.

* It can be also changed based on our preference and it is its major advantage.

ITERATIVE DEEPENING SEARCH [IDS]

IDS repeatedly performs depth-limited searches with increasing depth.

For example:

Depth 0 $\rightarrow$ Depth 1 $\rightarrow$ Depth 2 $\rightarrow$ Depth 3.

UCS & IDS for Robot Maze:

A robot in an unknown maze can use uninformed search strategies to find a path to the exit without knowing the maze layout beforehand.

1. Uniformed Cost Search (UCS):

UCS expands the path with the lowest total cost first.

Working:

1. Start from the robot's initial position.
2. Add possible paths to a priority queue.
3. Select the path with the lowest cost.
4. Continue until the exit is reached.

Advantages:

- * Requires relatively less memory.
- * Repeated Exploration can increase the time taken

Disadvantages:

- * Repeats searches at earlier depths.
- * Repeated Exploration can increase the time taken

Best Combinations:

The goal-based agent with IDS is suitable when memory is limited and search depth is unknown.

If different paths have different costs and robots need the least-cost path a goal-based agent with UCS is more suitable.

Conclusion:

The choice depends on one maze. IDS is useful when memory is limited while UCS is better when the aim is to find the least cost path.